

Improvement of the short channel effect in PMOSFETs using cold implantation



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ABSTRACT

In this paper, to suppress transient enhanced dopant diffusion and improve short channel effects, cold implantation (cold-IIP) was applied to contact PLUG implantation in P-channel metal oxide semiconductor field effect transistors (PMOSFETs). A shallow dopant profile was formed by the suppression of transient enhanced diffusion (TED) due to the reduction of end-of-range (EOR) defects. Threshold voltage roll-off and off current (I_{off}) increment, which are caused by a reduction in the distance between the gate and contact, were improved compared with room temperature implantation (RT-IIP). Additionally, the drain induced barrier lowering was improved, and the on-current improvement was attributed to reducing the contact resistance through the reduction of EOR defects. The contact resistance was reduced by ~6% of the RT-IIP. In the DRAM device, the standby current at a short propagation delay time (t_{PD}) was reduced effectively due to the decrease in the I_{off} and contact resistance for the cold-IIP case.

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1. Introduction

Short channel effects (SCE) such as the reduction of the threshold voltage (V_{TH} roll-off), degradation of the sub-threshold slope, and velocity saturation are limiting factors for device integration as devices are scaled down [1]. For improvements in the SCE, junction technology that reduces the junction depth has been studied and adopted to metal oxide semiconductor field effect transistor (MOSFET) fabrication. When producing a shallow junction, weak lateral spreading is desirable for control of SCE, while the source and drain resistances have to be as low as possible [2]. Cold implantation (cold-IIP) [3], germanium (Ge) pre-amorphization implantation (PAI) & carbon (C) co-implantation [4], and heavy dopant (indium to NMOS) implantation [5,6] were proposed for fabrication of the shallow junction. Ge PAI & C co-

implantation technology was used to suppress transient enhanced diffusion (TED) and to contain the diffusion of boron (B) or phosphorus (P) used to dope N/PMOS [4,7]. However, Ge PAI & C co-implantation causes the increment in the sheet resistance and junction leakage current because of additional Ge dopants and induced damage [8,9], and indium implantation increases the sheet resistance due to acceptor freeze-out effect [10]. Cold-IIP focuses on engineering damages caused during implantation and is used to increase the amorphous layer in place of Ge PAI [11,12]. In this paper, the effects of cold-IIP in terms of the amorphous layer thickness, dopant diffusion and device performance (V_{TH} , junction leakage current, and contact resistance) were investigated through transmission electron microscopy (TEM), secondary ion mass spectroscopy (SIMS) and electrical analyses.

2. Experiments

For the fabrication of a PMOSFET, arsenic (As) halo implantation was performed after the gate was etched and a lightly doped drain (LDD) region was formed. A sidewall spacer was then formed and S/D P⁺ (B) implantation was performed. B contact PLUG implantation was performed with the largest dose to form an ohmic contact. For the first part of the experiment, TEM analysis was conducted to

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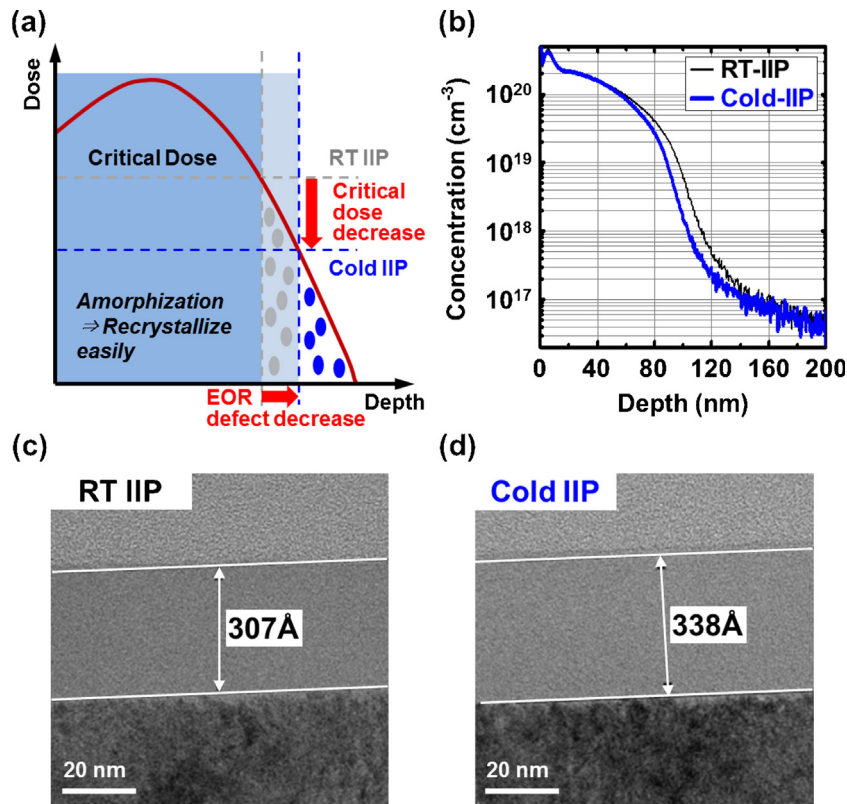


Fig. 1. (a) Schematic of the cold-IIP mechanism. When the cold-IIP technique is applied on Si, the amount of critical dose is decreased, thereby reducing the EOR defect area. (b) SIMS profile comparison between RT-IIP and cold-IIP. (c), (d) TEM images of RT-IIP and cold-IIP.

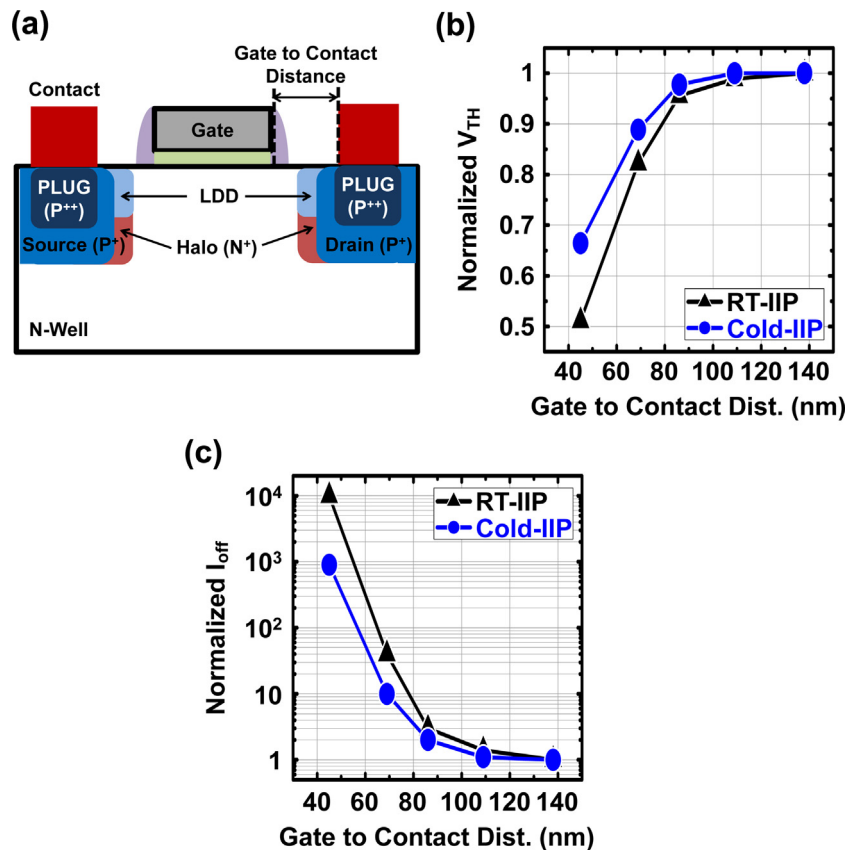


Fig. 2. (a) Schematic of the PMOS structure. (b) V_{TH} reduction according to reduction of the distance between the gate and the contact. (c) I_{off} increment according to shortening of the distance between the gate and the contact ((b) and (c) normalized with 140 nm).

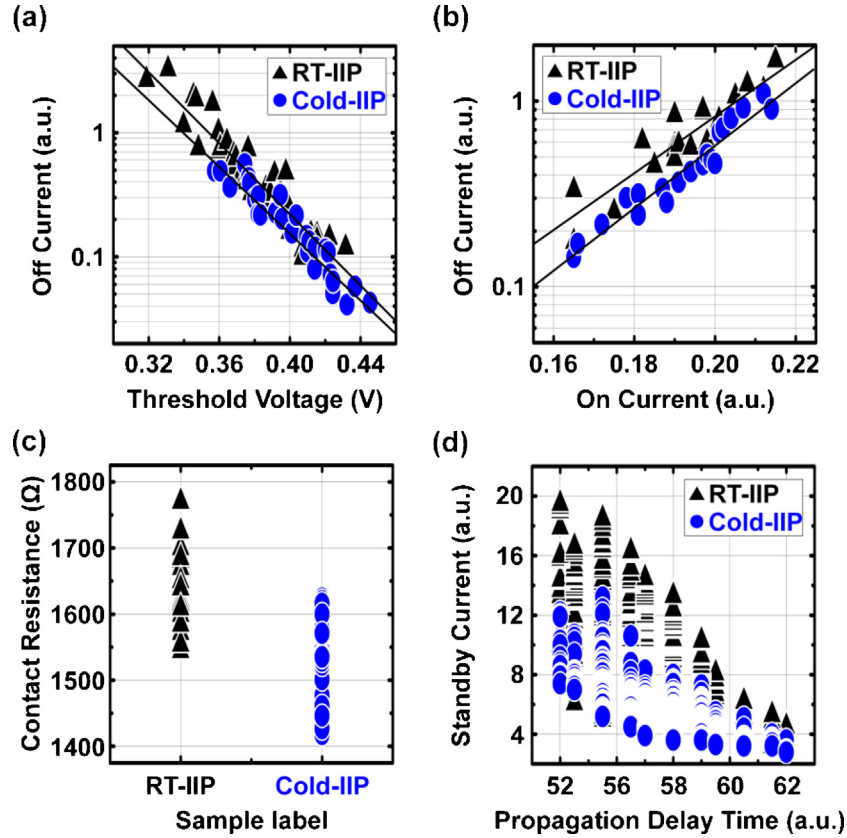


Fig. 3. (a) Correlation between V_{TH} and I_{off} . (b) Correlation between I_{on} and I_{off} . (c) Comparison of the contact resistance between RT-IIP and cold-IIP. (d) Standby current according to reduction of the propagation delay time.

compare the amorphous layer thickness for samples with RT-IIP (25 °C) and cold-IIP (−100 °C) when performing PLUG implantation. Additionally, we analyzed SIMS profiles after spike RTA at 1100 °C for 10 s to examine the profile change according to implantation temperature. RT-IIP was then performed in PLUG implantation and cold-IIP wafers while under the same processes. The V_{TH} and off current (I_{off}) values were measured and compared as the gate to the contact distance decreases for each RT-IIP and cold-IIP case. V_{TH} , on/off currents, contact resistance, and propagation delay time (t_{PD}) were measured for the whole wafer.

3. Results and discussion

Fig. 1(a) schematically shows the mechanism of cold implantation. Silicon (Si) lattices are interrupted by dopants during ion implantation and an amorphous layer is formed above the critical dose amount that acts as a threshold for the formation of the amorphous layer. If the implantation temperature is reduced, a self-annealing effect that elevates the lattice temperature and recrystallizes the amorphous layer is suppressed, the critical dose decreases [3], and the thickness of the amorphous layer is increased [13]. As shown in the cross sectional TEM images in Fig. 1(c) and (d), the amorphous layer thickness increased by approximately 30 Å in the cold-IIP sample. Additionally, at a low implantation temperature, the EOR defect area is reduced [14], which suppresses TED by Si interstitials [15]. Therefore, the diffusion of B is reduced and a shallow junction is eventually formed. From the SIMS profile shown in Fig. 1(b), a shallower profile was formed in the cold-IIP case than in the RT-IIP case.

As shown in the schematic of the PMOSFET structure at Fig. 2(a), contact PLUG implantation is performed for ohmic contact between S/D and the contact, and a dose greater than 10^{15} cm^{-2}

is required. In this case, heavily doped B is able to diffuse due to the TED effect and as the gate to contact distance is reduced, a V_{TH} roll-off phenomenon (PLUG effect) occurs. However, if cold-IIP is applied during PLUG implantation, TED is suppressed and the B diffusion is reduced. Therefore, as shown in Fig. 2(b), the V_{TH} for cold-IIP decreased less than that for RT-IIP. Additionally, because I_{off} is inversely proportional to V_{TH} , the I_{off} increased less in the cold-IIP case (Fig. 2(c)) [16].

Based on the results of the electrical measurement between RT-IIP and cold-IIP, V_{TH} is decreased by approximately 20 mV in the cold-IIP case at the same I_{off} value (Fig. 3(a)). Therefore, the effective channel length (L_{eff}) increased and the DIBL improved due to suppression of the B diffusion by cold-IIP [16]. Additionally, based on the correlation between I_{on} and I_{off} (Fig. 3(b)), I_{on} increased at the same I_{off} when using cold-IIP. The increase in I_{on} is attributed to a reduction in contact resistance through the reduction of EOR defects. As shown in Fig. 3(c), the contact resistance of cold-IIP was reduced by approximately 6% of the RT-IIP case. In the DRAM device, the standby current is a sum of several leakage currents and it increases as the speed of the transistor increases. The speed of the transistor was investigated through measurement of the propagation delay time (t_{PD}) at the inverter chain. In Fig. 3(d), the standby current at a short t_{PD} was reduced effectively due to the decrease in the I_{off} and contact resistance for the cold-IIP case.

4. Conclusion

By using cold-IIP, a shallow dopant profile was formed by the suppression of TED due to the reduction of EOR defects. When cold-IIP was applied to the contact PLUG implantation, V_{TH} roll-off and I_{off} increment according to the reduction of the distance

between the gate and contact was improved. Additionally, the DIBL was improved and the increase in the on current was attributed to reducing the contact resistance through the reduction of EOR defects. The contact resistance of cold-IIP was reduced by approximately 6% of the RT-IIP case. In the DRAM device, the standby current at a short t_{PD} was reduced effectively due to the decrease in the I_{off} and contact resistance for the cold-IIP case.

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